

LAC LA BICHE, AB, Canada

WMO: 713160

Lat: 54.7689N

Lon: 112.0236W

Elev: 567

StdP: 94.70

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.2	-38.1	0.1	-32.9	-34.8	0.1	-29.8	9.1	0.2	8.1	-1.9	1.0	170	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	27.5	17.3	25.7	16.4	24.1	15.6	18.8	25.2	17.7	23.6	16.6	22.1	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.5	21.6	15.3	11.6	20.2	14.3	10.9	19.1	55.8	25.3	52.0	23.8	48.7	22.1	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	6.2	-38.5	30.5	3.8	1.8	-41.2	31.8	-43.4	32.8	-45.5	33.8	-48.3	35.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-38.3	20.8	3.5	1.2	-40.8	21.7	-42.8	22.4	-44.8	23.1	-47.3	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-13.9	-11.3	-5.2	3.3	10.3	14.6	17.2	15.7	10.8	3.6	-5.6	-12.5
	DBStd	12.67	9.44	8.02	7.93	5.53	4.60	3.14	2.76	3.21	4.22	4.57	6.70	8.40
	HDD10.0	3484	742	597	470	208	53	4	0	2	41	202	467	699
	HDD18.3	5886	1000	831	728	451	252	118	55	92	228	456	717	957
	CDD10.0	677	0	0	0	7	61	141	223	178	64	4	0	0
	CDD18.3	37	0	0	0	0	2	5	19	11	1	0	0	0
	CDH23.3	513	0	0	0	2	42	89	209	142	30	1	0	0
	CDH26.7	86	0	0	0	0	7	13	40	22	4	0	0	0
Wind	WSAvg	2.7	2.4	2.5	2.8	3.1	3.0	2.7	2.7	2.5	2.8	2.9	2.6	2.4
Precipitation	PrecAvg	405	17	12	16	24	41	70	78	52	33	18	16	17
	PrecMax	507	39	27	34	57	86	138	155	90	74	56	41	39
	PrecMin	300	0	0	3	3	6	22	22	21	9	5	5	0
	PrecStd	61	11	7	9	13	19	30	27	19	18	11	9	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.0	11.8	22.2	27.4	28.3	30.1	29.0	26.7	20.0	10.5	7.4
		MCWB	2.8	2.4	5.1	10.8	14.3	16.4	19.3	18.8	16.0	11.2	5.3	2.6
	2%	DB	4.4	4.3	8.6	18.0	24.0	25.8	27.6	26.7	23.3	15.4	6.9	4.2
		MCWB	1.3	0.9	3.2	8.6	13.1	15.2	18.0	17.8	14.4	8.6	3.3	1.2
	5%	DB	2.1	2.0	6.7	15.0	21.6	23.8	25.6	24.8	20.8	13.0	4.7	1.5
		MCWB	-0.1	-0.7	2.1	6.9	11.4	14.4	17.2	16.7	13.3	7.2	1.8	-0.8
	10%	DB	-0.6	-0.1	4.7	12.2	19.3	21.8	23.9	22.9	18.3	10.8	2.6	-1.1
		MCWB	-2.5	-2.1	1.0	5.5	10.3	13.6	16.3	15.8	12.1	6.0	0.4	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.9	5.5	10.9	15.5	18.1	20.6	20.2	16.8	11.4	5.7	3.1
		MCDB	6.5	5.9	11.0	21.1	24.0	25.4	27.1	26.7	24.5	18.8	9.2	6.7
	2%	WB	1.6	1.1	3.6	8.9	13.7	16.6	19.3	18.6	15.1	9.2	3.8	1.2
		MCDB	4.1	4.0	8.0	17.0	21.9	22.8	25.6	24.9	21.9	14.4	6.5	4.1
	5%	WB	0.0	-0.5	2.4	7.2	12.4	15.5	18.1	17.4	13.9	7.7	2.0	-0.7
		MCDB	2.2	1.9	6.2	14.3	19.9	21.6	24.2	23.1	19.9	12.1	4.4	1.3
	10%	WB	-2.4	-2.2	1.2	5.7	11.1	14.5	17.1	16.3	12.6	6.4	0.5	-2.6
		MCDB	-0.7	0.0	4.4	11.6	17.9	20.2	22.4	21.6	17.4	10.2	2.4	-1.1
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.6	10.4	11.2	12.4	11.4	11.0	11.3	11.2	8.8	7.6	8.7
		MCDBR	11.0	12.4	11.8	15.7	16.3	15.1	14.1	14.6	15.8	13.5	10.0	11.0
		MCWBR	8.8	9.6	7.5	8.7	7.5	6.7	6.4	6.8	8.1	7.9	6.9	8.7
	5% WB	MCDBR	10.8	12.0	11.2	15.1	14.7	12.7	12.9	13.4	14.7	12.8	9.3	10.8
		MCWBR	8.7	9.5	7.3	8.6	7.1	6.3	6.5	6.7	7.7	7.7	6.7	8.8
Clear-Sky Solar Irradiance	taub	0.251	0.266	0.293	0.332	0.345	0.361	0.360	0.367	0.316	0.284	0.255	0.237	
	taud	2.295	2.325	2.369	2.388	2.403	2.367	2.430	2.352	2.511	2.576	2.457	2.287	
	Ebn at Noon	713	836	890	896	899	885	879	847	859	820	723	650	
	Edh at Noon	54	75	91	104	108	114	105	107	80	59	46	43	
All-Sky Solar Radiation	RadAvg	0.81	1.83	3.34	4.63	5.63	5.80	5.84	4.79	3.33	1.84	0.84	0.57	
	RadStd	0.09	0.14	0.25	0.37	0.51	0.40	0.32	0.36	0.35	0.28	0.10	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page